

Relevant information undersampling to support imbalanced data classification

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ABSTRACT

Traditional classification algorithms suppose that the sample distribution among classes is balanced. Yet, such an assumption leads to biased performance over the majority class. This paper proposes a Relevant Information-based UnderSampling (RIUS) approach to select the most relevant examples from the majority class to improve the classification performance for imbalanced data scenarios. RIUS builds on the information-preservation principle that extracts the majority class's underlying structure with fewer samples. Additionally, we couple our RIUS approach to the well-known Clustering-based Undersampling algorithm (CBUS) to enhance the data representation, and named this RIUS enhancement as CRIUS. Experimental results show that RIUS and CRIUS reveal the data's relevant structure and reduce the loss of information by selecting the most informative instances.

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1. Introduction

In machine learning applications, it is widespread to find datasets where one class has more samples than the other; this is known as the class imbalance problem. These kinds of applications are challenging to deal with since traditional classification algorithms usually assume that training samples are evenly distributed among classes. Without consideration of the imbalance problem, the classification algorithms can be overwhelmed by the majority class and can ignore the minority one [1]. This phenomenon is encountered in numerous real-world applications such as medical diagnosis [2], face recognition [3], detection of fraudulent transactions [4], among others.

Regarding this, there are three main alternatives to deal with the imbalanced data classification issue: i) *Algorithm-level approaches* that cope with the class distribution problem by modifying the learning stage. The most popular methodology is cost-sensitive learning [5]. A cost-sensitive classifier penalizes higher the samples from the minority class in comparison to those into the majority cluster. Nonetheless, misclassification costs are often unknown leading to overfitting [6]. ii) *Ensemble approaches*, which split the majority class into several subsets with equal size to the

number of instances in the minority group. Then, they train as many learning algorithms as the number of subgroups in the majority class [7]. Each classifier is trained separately, and the final decision is computed by majority voting. However, ensemble methods add learning time and memory constraints, i.e., Bagging and Boosting [8]. iii) *Data-level approaches* that adjust the class ratio in the input dataset to achieve balanced class distribution [9]. The latter includes two main strategies: oversampling and undersampling. The oversampling approaches augment the number of minority class samples to lessen the effects of the imbalance; notwithstanding, they increase the likelihood of overfitting [10]. On the contrary, the undersampling approaches reduce the number of samples from the majority class. The most common technique is Random-under-sampling (RUS). Nonetheless, some useful data of the majority class might be eliminated [11]. Then, smarter subsampling techniques have been developed. The clustering-based under-sampling (CBUS) technique performs the subsampling by partitioning the dataset in several clusters using the K-means algorithm, and replacing the original data by the centroids. Still, the K-means-based procedure is usually affected by outliers, and it is not robust to complex data distributions.

In this paper, we propose an undersampling approach based on a self-organizing principle of relevant information (PRI) [12,13]. Our relevant information-based undersampling approach, termed RIUS, allows representing the distribution of the majority class.

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RIUS selects the most informative instances of the majority cluster and reduces the loss of information in the undersampling process. The RIUS method discussed here is an extension of our work published in [14]. In particular, we coupled our PRI-based framework with a well-known clustering undersampling, developing a bagging-based ensemble that outperforms state-of-the-art methods in most tested datasets. The clustering-based enhancement of RIUS is called CRIUS. Besides, we extended RIUS's mathematical formulation and included both free parameter and computational time analysis. Additionally, we uploaded the Matlab implementation of both methods in a Github repository.

The remainder of this paper is organized as follows: Section 2 describes the related work. Section 3 provides the theoretical background. Sections 4 and 5 present the experiments and discuss the results obtained. Finally, Section 6 outlines the conclusions and the future work.

2. Related work

In recent years, several algorithms have been proposed to address the problem of imbalanced data classification. As stated in the introduction, these approaches can be classified into three categories, depending on how they cope with the problem: Algorithm-level, Ensembles, and Data-level strategies. Moreover, the data-level strategy can be embedded in ensemble learning algorithms to enhance system performance. The new hybrid method preprocesses the data (oversampling/undersampling) before training each classifier [8]. From an imbalanced dataset $\mathbf{X} \in \mathbb{R}^{N \times P}$ holding binary labels $z_n \in \{-1, +1\}$ ($n \in \{1, 2, \dots, N\}$), we can define the matrix $\mathbf{X}_+ \in \mathbb{R}^{N_+ \times P}$ gathering samples belonging to the majority class, $z = +1$, and $\mathbf{X}_- \in \mathbb{R}^{N_- \times P}$ as the matrix of training data belonging to the minority one, $z = -1$, where $N_+ \gg N_-$ and $N = N_+ + N_-$. Data-level approaches and hybrid methods imposed a balanced distribution, which means that $N_- \equiv N_+$.

Oversampling methods are mainly ruled by the Random Oversampling approach that replicates minority class instances. Still, it can increase the overfitting because it makes exact copies of existing observations [8]. Also, ensemble-based approaches have been proposed through bagging, i.e., Overbagging [15]. Besides, the synthetic minority oversampling technique, known as SMOTE, is thriving in a variety of applications involving imbalanced databases [16]. The SMOTE algorithm creates artificial data between samples of the minority class. Specifically, for the matrix $\mathbf{X}_-$, it considers the K nearest neighbors of each sample. Next, SMOTE creates a synthetic instance by randomly selecting a neighbor; then, the corresponding difference vector is multiplied by a random number between the range $[0, 1]$, and finally, the previous result is added to the vector. Although it has shown positive results, the SMOTE algorithm also has drawbacks, including over-generalization and convergence variability [17]. SMOTE is also combined with bagging and boosting-based ensembles, i.e., SMOTEBagging [15], and SMOTEBoost [18]. These approaches usually achieve competitive performance; but, the computational requirements are often enormous [19]. Also, they are prone to over-fitting, and the tuning of the free parameters is not a straightforward task [20].

Regarding the undersampling approaches, the well-known Random Undersampling method balances the class distribution through random elimination. The main drawback is that some useful samples in the majority class might be discarded [11]. This method has been coupled with different kinds of ensemble approaches. UnderBagging proposed by authors in [21] is a combination of a Random Undersampling process and a bagging-based ensemble. Likewise, Seiffert et al. [22], proposed RUSBoost, combining the Random Undersampling approach with boosting. In this sense, the boosting modifies the weight's distribution used to train the classifier toward the minority class and removes instances

from the majority cluster. Moreover, authors in [23] proposed a collection of undersampling methods named Near Miss, which selects examples based on a distance between the majority and minority classes. Tomek-link, proposed in [24], is a refinement of the Condensed Nearest Neighbor technique that eliminates boundary instances without harming the general data structure. Another representative undersampling algorithm is the Edited Nearest Neighbor technique [25], which removes an example from the majority class if the number of neighbors from the minority one is predominant. Nonetheless, the algorithms mentioned above are based on elementary rules founded on the Euclidean distance and simple random sampling, yielding poor representations of the underlying data structure.

Clustering-based undersampling methods aim to code the data structure before sampling the majority class. Yen et al. in [6], propose to cluster all training samples into K disjoint groups. Afterward, the approach randomly selects an adequate number of instances from each group, considering the imbalanced data ratio. Notwithstanding, this method requires the empirical tuning of the number of clusters to create. A new variant was recently proposed by authors in [10]. In this method, the majority class is grouped into N_- clusters, where N_- is the number of minority class samples. Then, the original data in the same groups are replaced by the cluster centroids. Additionally, it has been combined with Adaboost to create a hybrid approach.

Summarizing, data-level approaches devoted to imbalanced classification do not correctly code the relevant underlying data structure. They lack a suitable representation beyond second-order statistics (Euclidean distance) and employ random data selection without highlighting relevant samples from the majority class.

3. Methods

3.1. Information theoretic learning fundamentals

Let X be a random variable with probability density function (PDF) $f(x)$, where $x \in \mathcal{X}$ and $\mathcal{X}$ is a finite set. Information theoretic learning (ITL) relies on the well-known Renyi's α -order entropy $H_\alpha(X) \in \mathbb{R}$ defined as [26]:

$$H_\alpha(X) = \frac{1}{1-\alpha} \log \left(\int_{\mathcal{X}} f^\alpha(x) dx \right). \quad (1)$$

The limiting case $\alpha \rightarrow 1$ yields Shannon's entropy [27], and when $\alpha = 2$, the quadratic Renyi's entropy, $H_2(X) = -\log \left(\int_{\mathcal{X}} f^2(x) dx \right)$, provides a straightforward estimator.

Given a set of i.i.d. samples $\{x_n \in \mathcal{X}\}_{n=1}^N$ and a Gaussian kernel function $\kappa_\sigma : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, we can approximate the PDF through the Parzen estimator as follows:

$$\hat{f}(x) = \frac{1}{N} \sum_{n=1}^N \kappa_\sigma(x, x_n) = \frac{1}{N} \sum_{n=1}^N \exp \left(\frac{-\|x - x_n\|_2^2}{2\sigma^2} \right); \quad (2)$$

where $\sigma \in \mathbb{R}^+$ is a bandwidth and $\|\cdot\|_2$ stands for the L2 norm. Hence, the $H_2(X)$ can be approximated as $\hat{H}_2(X) = -\log(V(X))$, where $V(X) \in \mathbb{R}^+$ is known as the information potential (IP) defined as:

$$V(X) = \frac{1}{N^2} \sum_{n=1}^N \sum_{m=1}^N \kappa_{\sqrt{2}\sigma}(x_n, x_m). \quad (3)$$

Moreover, ITL allows us to compute directly a notion of distance between PDFs from data based on the Cauchy-Schwarz (CS) divergence. Let $\{x_n \in \mathcal{X}\}_{n=1}^N$ and $\{y_m \in \mathcal{Y}\}_{m=1}^M$ be a pair of sets of i.i.d samples drawn from the random variables X and Y , holding PDF $f(x)$ and $\xi(y)$, respectively. The CS divergence between X and Y is given by:

$$D_{CS}(X, Y) = 2H_2(X, Y) - H_2(X) - H_2(Y), \quad (4)$$

where the quadratic Renyi's cross-entropy $H_2(X, Y)$ in Eq. 4 can be estimated as: $\hat{H}_2(X, Y) = -\log(V(X, Y))$. The term $V(X, Y)$ is known as the cross-information potential defined as follows:

$$V(X, Y) = \frac{1}{NM} \sum_{n=1}^N \sum_{m=1}^M \kappa_{\sqrt{2}\sigma}(x_n, y_m). \quad (5)$$

In the following sections, the Renyi's quadratic entropy and the CS divergence will be used to select relevant samples within an imbalanced data classification problem.

3.2. Relevant information-based undersampling-(RIUS)

In a binary imbalanced data classification scenario, a discriminating function $g: \mathcal{X} \rightarrow \mathcal{Y}$ is learned from a set $\{\mathbf{x}_n, z_n\}_{n=1}^N$, where $\mathbf{x}_n \in \mathcal{X} \subseteq \mathbb{R}^P$ is a P -dimensional input feature vector corresponding to the n -th instance with output label $z_n \in \{-1, +1\}$. We define $\mathbf{X}_+ \in \mathbb{R}^{N_+ \times P}$ as the matrix of training data belonging to the majority class, $z = +1$, and $\mathbf{X}_- \in \mathbb{R}^{N_- \times P}$ as the matrix of training data belonging to the minority one, $z = -1$, where $N_+ \gg N_-$ and $N = N_+ + N_-$. We aim to reveal the M most representatives samples from the majority class, such that $M < N_+$, to prevent biased classification results. Therefore, we introduce a self-organizing principle of relevant information (PRI) in the context of imbalanced classification to capture data's underlying structure. Namely, we are looking for the subset $\mathbf{X}_+ \in \mathbb{R}^{M \times P}$ that minimizes the following PRI-based cost function [12]:

$$J(\mathbf{X}_+) = H_2(\mathbf{X}_+) + \lambda D_{CS}(\mathbf{X}_+, \mathbf{X}_+), \quad (6)$$

where $\lambda \in \mathbb{R}^+$ is a trade-off parameter that controls the level of distortion in the subsampled data. As seen in Eq. 6, we measure the majority class redundancy using the Renyi's quadratic entropy $H_2(\mathbf{X}_+)$ and quantify the distortion between the majority class and the processed (undersampled) data by the CS divergence $D_{CS}(\mathbf{X}_+, \mathbf{X}_+)$. Rewriting Eq. 6 concerning the IP and the CIP, the following optimization arises:

$$\mathbf{X}_+^* = \arg \min_{\mathbf{X}_+} (\lambda - 1) \log(V(\mathbf{X}_+)) - 2\lambda \log(V(\mathbf{X}_+, \mathbf{X}_+)). \quad (7)$$

Now, differentiating with respect to $\mathbf{x}_{+k} \in \mathbf{X}_+$ ($k \in \{1, 2, \dots, M\}$):

$$\frac{\partial J(\mathbf{X}_+)}{\partial \mathbf{x}_{+k}} = \frac{2(1-\lambda)}{V(\mathbf{X}_+)} \frac{\partial V(\mathbf{X}_+)}{\partial \mathbf{x}_{+k}} + \frac{2\lambda}{V(\mathbf{X}_+, \mathbf{X}_+)} \frac{\partial V(\mathbf{X}_+, \mathbf{X}_+)}{\partial \mathbf{x}_{+k}}. \quad (8)$$

According to Eq. 8, samples in $\mathbf{X}_+$ move under the influence of two main forces: an information force $F(\mathbf{x}_{+k}, \mathbf{X}_+) = \partial V(\mathbf{X}_+)/\partial \mathbf{x}_{+k}$ exerted by all samples in the subset $\mathbf{X}_+$ and a cross-information force $F(\mathbf{x}_{+k}, \mathbf{X}_+) = \partial V(\mathbf{X}_+, \mathbf{X}_+)/\partial \mathbf{x}_{+k}$ applied by samples belonging to the majority class in $\mathbf{X}_+$. Finally, rearranging Eq. 8 and fixing a Gaussian kernel to approximate the IP-based quantities (see Eq. 3.5) gives us the following fixed-point rule:

$$\begin{aligned} \mathbf{x}_{+k}^{(t+1)} = & \varsigma \frac{(1-\lambda)}{\lambda} \frac{\sum_{m=1}^M \kappa_{\sqrt{2}\sigma}(\mathbf{x}_{+k}^{(t)}, \mathbf{x}_{+m}^{(t)}) \mathbf{x}_{+m}^{(t)}}{\sum_{n=1}^{N_+} \kappa_{\sqrt{2}\sigma}(\mathbf{x}_{+k}^{(t)}, \mathbf{x}_{+n}^{(t)})} + \frac{\sum_{n=1}^{N_+} \kappa_{\sqrt{2}\sigma}(\mathbf{x}_{+k}^{(t)}, \mathbf{x}_{+n}^{(t)})}{\sum_{n=1}^{N_+} \kappa_{\sqrt{2}\sigma}(\mathbf{x}_{+k}^{(t)}, \mathbf{x}_{+n}^{(t)})} \mathbf{x}_{+n}^{(t)} \\ & - \varsigma \frac{(1-\lambda)}{\lambda} \frac{\sum_{m=1}^M \kappa_{\sqrt{2}\sigma}(\mathbf{x}_{+k}^{(t)}, \mathbf{x}_{+m}^{(t)})}{\sum_{n=1}^{N_+} \kappa_{\sqrt{2}\sigma}(\mathbf{x}_{+k}^{(t)}, \mathbf{x}_{+n}^{(t)})} \mathbf{x}_{+k}^{(t)}, \end{aligned} \quad (9)$$

where $\varsigma = (N_+ V(\mathbf{X}_+, \mathbf{X}_+)) / (M V(\mathbf{X}_+))$ and $t \in \mathbb{N}$ stands for the t -th iteration. Our Relevant Information-based Undersampling approach, termed RIUS, can be summarized as in Algorithm 1.

Algorithm 1: Relevant information undersampling-(RIUS)

Require: $\mathbf{X}_+$, σ and λ .

Ensure: Subset $\mathbf{X}_+$.

- 1: Generate randomly M samples belonging to $\mathbf{X}_+^{(0)}$.
 - 2: Compute the IP of $\mathbf{X}_+$ (Eq. 3).
 - 3: Compute the CIP between $\mathbf{X}_+$ and $\mathbf{X}_+$ (Eq. 5).
 - 4: Compute the cost function in Eq. 7.
 - 5: Update the samples (particles) in $\mathbf{X}_+$ as in Eq. 9.
 - 6: Iterate steps 2 through 5 until convergence.
-

3.3. Clustering-based RIUS enhancement-(CRIUS)

An essential drawback concerning our RIUS approach is that the IP and CIP estimations require $\mathcal{O}(M^2)$ and $\mathcal{O}(NM)$ operations, respectively, which may create computation bottlenecks for large datasets not mentioning the imbalanced issues. We propose to couple our RIUS approach to the well-known clustering-based undersampling algorithm (CBUS) to overcome these limitations. The pipeline of our Clustering-based enhancement of RIUS-(CRIUS) is shown in Fig. 1. We separate the training partition of the imbalanced dataset into the majority and the minority classes. Then, we divide the majority class into K disjoint groups based on the well-known K-means algorithm. We set the number K as equal to the number of minority class samples, which is $K = N_-$ [10]. Next, we perform the RIUS algorithm to select the M most relevant samples from each cluster.

There are two principal motivations for using the CBUS technique in combination with the RIUS approach. First, clustering the data into $K = N_-$ clusters implies that each group will have, on average, as many samples as the imbalance order; then, the RIUS constraint of the number of data becomes not problematic. Secondly, the original CBUS selects the centroid of each group. Still, the mean can be biased if the cluster has outliers or if its shape is non-Gaussian. In this sense, we propose choosing instead of the centroid, $M \geq 1$ samples from each group. Thus, the majority class will have $N_+ = M \times N_-$ instances, which means that the dataset will be balanced at $M : 1$. This procedure allows us to employ a bagging ensemble approach by training M (odd) classifiers; each of them trained on all the minority class instances and M equal-sized samples from the subsampled majority set. Finally, the classification of new observations is based on the aggregation of predictions from all the classifiers.

For the sake of clarity, let us suppose we have a dataset with 220 examples, 200 belonging to the majority class, and 20 in the minority one. To run the CRIUS algorithm, we cluster the majority class in 20 groups (the number of minority class samples). Then, we would have 20 clusters holding ten observations each. Next, we perform the RIUS algorithm over each group to select M samples. If we set $M = 3$, we choose three examples on each cluster and $N_+ = M \times N_- = M \times K = 3 \times 20 = 60$ observations in the undersampled majority class. In this way, the dataset will be balanced at order $M : 1 = 60 : 20 = 3 : 1$. In turn, we can perform a bagging ensemble approach by training three classifiers, each one of them trained on all the minority class instances and three equal-sized samples from the subsampled majority set.

It is worth mentioning how the CRIUS implementation changes for small imbalance ratios, i.e., $IR < 4$. In these cases, most clusters selected would have less than four samples, being necessary to duplicate (upsample) some regions of the input space to fulfill the required IR. However, the introduced ITL-based strategy will reproduce some observations following the cluster's relevant data

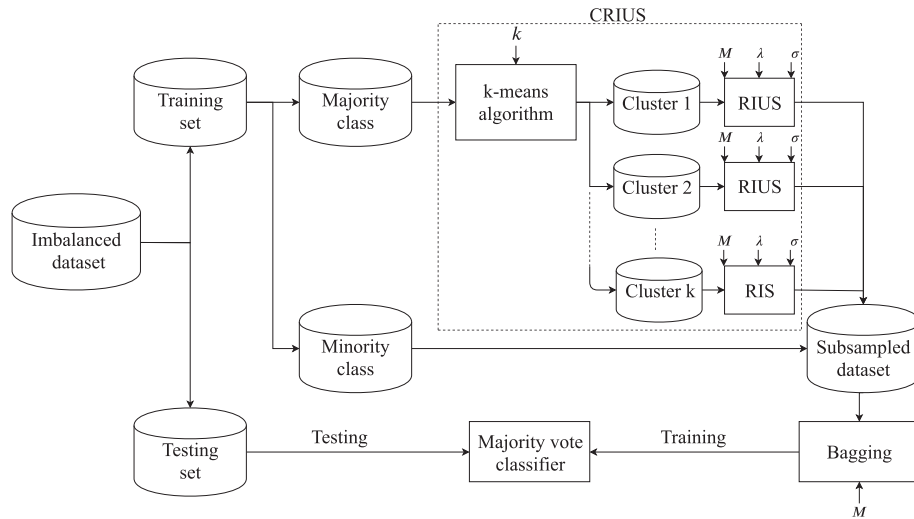


Fig. 1. CRIUS framework to support imbalanced data classification.

structure. The latter does not negatively impact the CRIUS's bagging classification stage. Though the instances can be duplicated, the bagging framework will be trained with equal-sized data from the majority class's relevant structure.

4. Experimental setup

4.1. Datasets

To evaluate our RIUS and CRIUS approaches as undersampling techniques to deal with imbalanced binary classification, we perform experiments on synthetic and real-world datasets. A representative analysis is carried out to compare the RIUS and CRIUS functioning with two conventional subsampling approaches. A 2D sample space of points distributed in a non-linear arrangement of three clusters is considered (see Fig. 2). Also, we analyze the free parameters on two synthetic datasets: i) two groups of points chosen from two bivariate normal distributions with different means ($\mu_1 = [-1.5, -1.5]$ and $\mu_2 = ([1.5, 1.5])$), moderately overlapped; and ii) the well-known two half-moons database. For the real-world experiments, we use 44 small-scale datasets for binary classification from the Knowledge extraction based on evolutionary learning (KEEL) repository [8].¹ KEEL holds datasets with imbalance ratios varying between 1.8 and 129, and the number of samples varying between 130 and 5500. A brief description is depicted in Table 1. Moreover, the proposed approach is also tested on two large-scale binary databases from the Knowledge Discovery and Data Mining Cup (KDDMC),² which contain 102294 and 145751 samples, 117 and 74 features, and imbalance ratios of 163 and 111, respectively.

4.2. RIUS and CRIUS training

Before performing the undersampling, we employ a z-score normalization for all training data. In turn, to implement the RIUS algorithm, we take the majority class $\mathbf{X}_+$ of each dataset and place M samples belonging to $\mathbf{X}_+$ at random locations within a P -dimensional hypercube in the range between -2 and 2 . After normalization, we expect to find approximately the 95% of data. The M value is set equal to the number of minority class samples (i.e., $M = N_-$); consequently, both the majority and minority

classes contain the same amount of data. Afterward, we carried out RIUS, allowing the particles to self-organize. Regarding the CRIUS, the M parameter was set as $M = 3$ for the real-world datasets to employ a bagging ensemble approach by training three classifiers (the final decision is made by majority voting). To examine the classification performance, we implement an RBF Support Vector Machine (SVM) classifier. A 5-fold nested cross-validation strategy tunes the SVM's free parameters. For concrete testing, the scale value is searched within the logspace range $[0.01\sigma_0, 3\sigma_0]$, where σ_0 is the median of the Euclidean distance among input samples, and the penalty value is searched within the range $\{1, 2, 5, 10, 100, 500\}$. We assess the proposed methods' classification performance by measuring the Area Under the ROC Curve (AUC), the Geometric Mean (GM), the sensitivity, and the specificity. The data partitions used in this paper can be found in the repositories above described so that any interested researcher can reproduce the experimental study. Our Matlab implementation of RIUS and CRIUS is also uploaded in a Github repository.³

4.3. Method comparison

To compare RIUS and CRIUS performances, we chose state-of-the-art approaches that fulfill the following facts: i) The technique has proven to outperform other state-of-the-art strategies for dealing with imbalanced datasets [8,10]. ii) The algorithm consists of a combination of undersampling and ensemble learning. iii) The method must be tested on the KEEL (small scale datasets) and/or the KDDMC (large scale datasets) repositories. Then, the studied approaches include: i) *RUSBoost1* (RUS1) [22,28]. This approach is a combination of RUS (random undersampling) and the standard boosting procedure AdaBoost. ii) *Underbagging4* (UB4) [21,29]. It is a combination of RUS and a bagging procedure, which does not require recomputing weights as AdaBoost. iii) *SMOTEBagging4* (SBAG4) [15,30]. This method involves the generation step of synthetic instances from minority class (SMOTE) during subset construction in the bagging procedure. iv) *Clustering-based undersampling with AdaBoost* (CUS-AB) [10]. It splits the majority class into K clusters, where the number of groups is set to be equal to the number of samples in the minority class (i.e., $K = N_-$). Then, a nearest neighbor-based strategy is used to replace the majority class to apply AdaBoost. It is worth mentioning that the original method and name abbreviations are used. We append a "1" to

¹ <http://www.keel.es/dataset.php>.

² <http://www.kdd.org/kdd-cup>.

³ <https://github.com/keider95/Relevant-Information-Sampling/>.

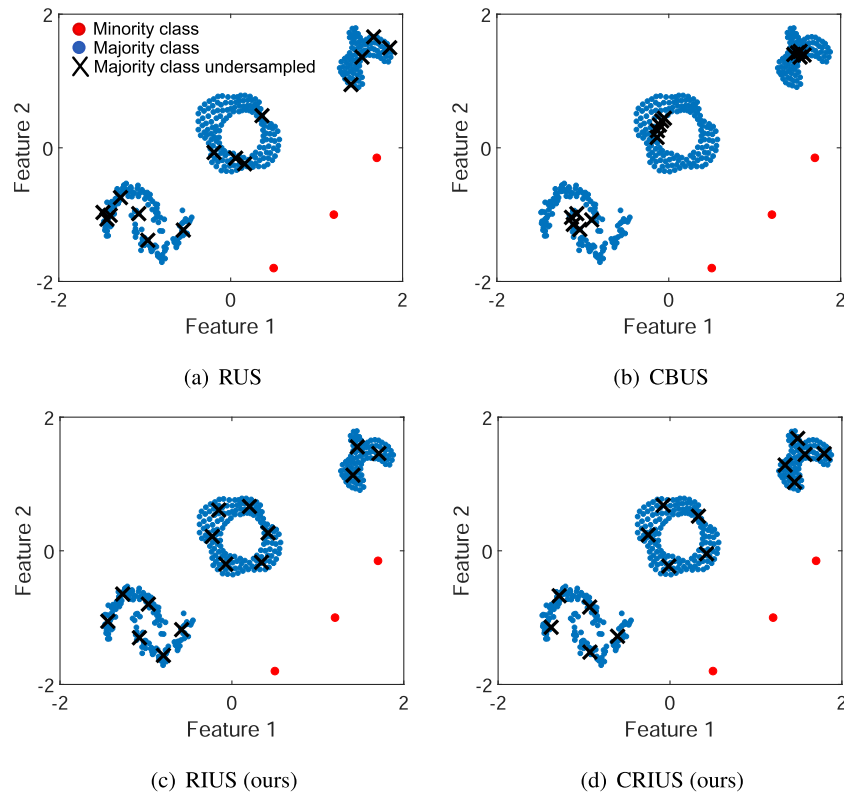


Fig. 2. 2D synthetic experiment - illustrative results. RIUS and CRIUS parameters are fixed as: $\lambda = 100$ and $\sigma = 0.5$. (a) Random undersampling-RUS. (b) Clustering-based undersampling-CBUS. (c) RIUS (ours), $M = 15$. (d) CRIUS (ours), $K = 3$, $M = 5$.

Table 1

KEEL dataset description. N : Number of samples, P = Number of features, IR: Imbalance ratio.

	Dataset	N	P	IR		Dataset	N	P	IR
1	Abalone9-18	731	8	16.68	23	Page-blocks13vs2	472	10	15.85
2	Abalone19	4174	8	128.87	24	Pima	768	8	1.9
3	Ecoli-0_vs_1	220	7	1.86	25	Segmemt0	2308	19	6.01
4	Ecoli-0-1-3-7_vs_2-6	281	7	39.15	26	Shuttle0vs4	1829	9	13.87
5	Ecoli1	336	7	3.36	27	Shuttle2vs4	129	9	20.5
6	Ecoli2	336	7	5.46	28	Vehicle0	846	18	3.23
7	Ecoli3	336	7	8.19	29	Vehicle1	846	18	2.52
8	Ecoli4	336	7	13.84	30	Vehicle2	846	18	2.52
9	Glass0	214	9	3.19	31	Vehicle3	846	18	2.52
10	Glass0123vs456	192	9	10.29	32	Vowel0	988	13	10.1
11	Glass016vs2	184	9	19.44	33	Wisconsin	683	9	1.86
12	Glass016vs5	214	9	1.82	34	Yeast05679vs4	528	8	9.35
13	Glass1	214	9	10.39	35	Yeast1	1484	8	2.46
14	Glass2	214	9	15.47	36	Yeast1vs7	459	8	13.87
15	Glass4	214	9	22.81	37	Yeast1289vs7	947	8	30.56
16	Glass5	214	9	22.81	38	Yeast1458vs7	693	8	22.1
17	Glass6	214	9	6.38	39	Yeast2vs4	514	8	9.08
18	Haberman	306	3	2.68	40	Yeast2vs8	482	8	23.1
19	Iris0	150	4	2	41	Yeast3	1484	8	8.11
20	New-thyroid1	215	5	5.14	42	Yeast4	1484	8	28.41
21	New-thyroid2	215	5	4.92	43	Yeast5	1484	8	32.78
22	Page-blocks0	5472	10	8.77	44	Yeast6	1484	8	39.15

the algorithm's abbreviation highlighting that it employs ten classifiers, and we do the same with a "4" whenever it uses 40.

5. Results and discussions

5.1. RIUS and CRIUS illustrative results

As an illustrative experiment, Fig. 2 shows the undersampling results obtained over the 2D synthetic data holding clusters with

a non-linear arrangement. We aim to balance at order 5:1 but preserving the global data structure. As seen, RUS selects points randomly without consideration of the data shape, and the general structure is ignored. Of note, the RUS approach does not preserve the global structure of the majority class (see Fig. 2 (a)). On the other hand, Fig. 2 (b) shows how CBUS groups the majority class into three clusters, and it selects five samples from each group; in this case, the five nearest neighbors to the cluster center are chosen. Also, note that the centroid is not part of the original samples; therefore, they do not represent it correctly.

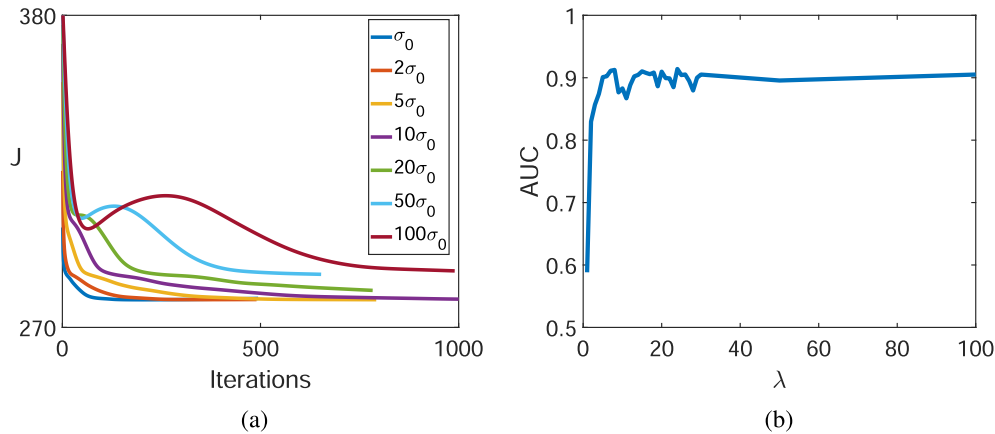


Fig. 3. RIUS free parameter analysis over synthetic data composed of two bivariate normal distributions with different means and overlapping. (a) RIUS cost function in terms of different σ values vs. iterations, and (b) AUC vs. λ .

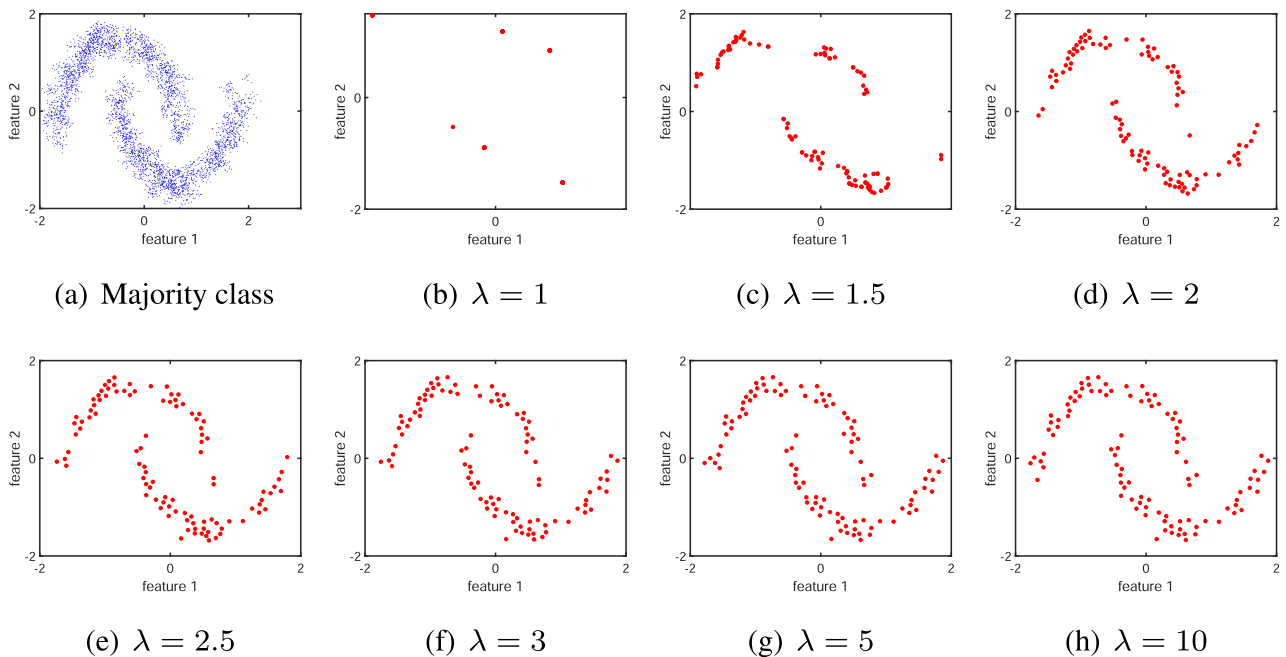


Fig. 4. Illustration of the impact of different λ values in RIUS behavior. Blue points: majority class, Red points: subsampled majority class. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Fig. 2(c) and (d) show the samples selected by RIUS and CRIUS techniques. RIUS seeks the 15 most relevant examples from the entire majority class. Note that some clusters have different instances that prioritize the middle and bottom groups and place just three observations on the upper group. CRIUS selects the five most relevant samples from each group, obtaining the same number of points from each data conglomerate, which for classification purposes, could become an advantage since it guarantees that for each data structure, the same amount of samples are selected.

To figure out the influence of the RIUS and CRIUS free parameters, it is worth mentioning that two key variables rule their performance: the weighting value λ and the bandwidth σ . The first one allows combining the marginal entropy and the divergence in Eq. 6; meanwhile, the second one scales the L2 norm of the Gaussian kernel in Eq. 2. For the bandwidth, we implement the ITL-based strategy proposed by authors in [12], which reduces the risk of being trapped in a local minimum and makes the algo-

rithm flexible and robust. The learning strategy consists of allowing the kernel size to decrease over iterations according to $\sigma_t = \frac{\sigma}{1+\gamma t}$, where $\gamma \in \mathbb{R}^+$ is a decaying rate. To find the appropriate values for σ and γ , we perform some synthetic experiments, as discussed in Section 4. Consequently, we fix $\gamma = 1$ and evaluate seven different kernel sizes: $\sigma \in \{\sigma_0, 2\sigma_0, 5\sigma_0, 10\sigma_0, 20\sigma_0, 50\sigma_0, 100\sigma_0\}$. Fig. 3(a) shows the cost function and the number of iterations in which the method converges for each kernel scale value. We can see that the smaller the σ value, the faster the algorithm converges. This finding demonstrates that it is appropriate to establish $\sigma = \sigma_0$ because it ensures a more rapid and stable convergence. However, it is essential to highlight that a σ value smaller than σ_0 can make the algorithm unstable and prevent it from converging.

Concerning λ , we compare the solutions reached by different values of this parameter on the two-Gaussians dataset. Comparing the majority class's subsampled versions with respect to the AUC and the λ values can be observed in Fig. 3(b). We can see that for

Table 2

RIUS and CRIUS classification results. KEEL (small scale datasets) and KDDMC (large scale dataset) repositories are studied. The sensitivity, specificity, G-mean, and AUC of the testing set is presented. Bold: Best results per database.

Dataset	Sensitivity				Specificity				G-mean				AUC			
	RIUS		CRIUS		RIUS		CRIUS		RIUS		CRIUS		RIUS		CRIUS	
Small scale datasets																
Abalone9-18	0,88	0,13	0,93	0,06	0,73	0,03	0,83	0,03	0,80	0,07	0,88	0,03	0,89	0,37	0,96	0,03
Abalone19	0,81	0,17	0,69	0,16	0,68	0,06	0,75	0,02	0,73	0,06	0,71	0,08	0,81	0,07	0,82	0,08
Ecoli-0_vs_1	0,97	0,03	0,98	0,03	0,97	0,04	0,97	0,04	0,95	0,00	0,95	0,03	0,98	0,02	1,00	0,00
Ecoli-0-1-3-7_vs_2-6	1,00	0,00	0,70	0,45	0,50	0,27	0,86	0,19	0,69	0,19	0,67	0,39	0,84	0,24	0,95	0,08
Ecoli1	0,96	0,06	0,94	0,11	0,75	0,09	0,83	0,06	0,85	0,06	0,88	0,06	0,93	0,05	0,95	0,07
Ecoli2	0,93	0,08	0,93	0,08	0,90	0,10	0,95	0,03	0,91	0,05	0,94	0,04	0,96	0,03	0,94	0,04
Ecoli3	0,89	0,16	0,89	0,19	0,81	0,06	0,88	0,06	0,84	0,05	0,88	0,09	0,94	0,02	0,94	0,02
Ecoli4	0,95	0,11	0,90	0,14	0,82	0,17	0,97	0,02	0,87	0,09	0,93	0,07	0,98	0,02	0,99	0,02
Glass0	0,81	0,14	0,89	0,11	0,78	0,04	0,71	0,10	0,79	0,08	0,79	0,07	0,88	0,06	0,87	0,05
Glass0123vs456	1,00	0,00	0,90	0,07	0,85	0,06	0,95	0,04	0,92	0,03	0,92	0,03	0,97	0,03	0,98	0,01
Glass016vs2	0,68	0,32	0,70	0,24	0,67	0,16	0,52	0,17	0,60	0,38	0,58	0,14	0,80	0,15	0,84	0,10
Glass016vs5	1,00	0,00	0,80	0,27	0,55	0,18	0,86	0,14	0,73	0,12	0,82	0,20	0,91	0,07	0,96	0,08
Glass1	0,82	0,17	0,79	0,11	0,64	0,09	0,64	0,11	0,68	0,06	0,71	0,02	0,71	0,06	0,81	0,06
Glass2	0,85	0,22	0,88	0,16	0,69	0,05	0,45	0,16	0,72	0,10	0,62	0,10	0,86	0,10	0,82	0,15
Glass4	1,00	0,00	0,73	0,43	0,63	0,24	0,94	0,06	0,78	0,16	0,73	0,42	0,94	0,07	0,97	0,03
Glass5	1,00	0,00	0,80	0,27	0,55	0,22	0,92	0,09	0,73	0,16	0,84	0,14	0,91	0,10	0,98	0,03
Glass6	1,00	0,00	0,82	0,14	0,72	0,03	0,97	0,02	0,85	0,02	0,89	0,08	0,97	0,03	0,98	0,02
Haberman	0,57	0,14	0,62	0,07	0,68	0,04	0,61	0,10	0,62	0,08	0,61	0,08	0,69	0,02	0,60	0,07
Iris0	0,98	0,04	0,98	0,04	1,00	0,00	1,00	0,00	0,99	0,02	0,99	0,02	1,00	0,00	1,00	0,00
New-thyroid1	1,00	0,00	1,00	0,00	0,82	0,09	0,97	0,03	0,90	0,05	0,98	0,02	0,99	0,03	1,00	0,00
New-thyroid2	1,00	0,00	1,00	0,00	0,76	0,08	0,96	0,03	0,87	0,05	0,98	0,02	0,98	0,03	1,00	0,00
Page-blocks0	0,96	0,02	0,95	0,03	0,90	0,03	0,94	0,01	0,93	0,01	0,95	0,02	0,97	0,00	0,98	0,00
Page-blocks13vs4	1,00	0,00	1,00	0,00	0,91	0,03	0,93	0,03	0,95	0,01	0,96	0,02	0,99	0,01	1,00	0,01
Pima	0,81	0,04	0,76	0,08	0,66	0,04	0,68	0,05	0,73	0,02	0,72	0,04	0,82	0,03	0,81	0,04
Segment0	1,00	0,00	1,00	0,00	1,00	0,00	1,00	0,00	1,00	0,00	1,00	0,00	1,00	0,00	1,00	0,00
Shuttle0vs4	1,00	0,00	0,99	0,02	1,00	0,00	1,00	0,00	1,00	0,00	1,00	0,01	1,00	0,00	1,00	0,00
Shuttle2vs4	0,90	0,22	0,90	0,22	0,94	0,14	1,00	0,00	0,91	0,13	0,94	0,13	1,00	0,00	1,00	0,00
Vehicle0	1,00	0,01	1,00	0,00	0,95	0,02	0,97	0,02	0,97	0,01	0,98	0,01	0,99	0,01	1,00	0,00
Vehicle1	0,91	0,03	0,88	0,04	0,75	0,04	0,82	0,02	0,83	0,04	0,84	0,02	0,91	0,03	0,92	0,02
Vehicle2	0,97	0,03	0,97	0,03	0,98	0,01	0,97	0,01	0,98	0,01	0,97	0,02	0,99	0,01	1,00	0,01
Vehicle3	0,86	0,05	0,88	0,07	0,69	0,08	0,78	0,03	0,77	0,05	0,83	0,02	0,87	0,02	0,91	0,02
Vowel0	0,96	0,07	0,96	0,05	0,98	0,02	0,99	0,01	0,97	0,03	0,97	0,02	1,00	0,00	1,00	0,00
Wisconsin	0,99	0,01	0,97	0,02	0,95	0,02	0,95	0,02	0,97	0,01	0,96	0,02	0,98	0,01	0,99	0,01
Yeast05679vs4	0,84	0,12	0,73	0,15	0,64	0,10	0,80	0,03	0,73	0,05	0,76	0,08	0,83	0,05	0,87	0,05
Yeast1	0,69	0,06	0,82	0,04	0,64	0,07	0,57	0,02	0,66	0,03	0,68	0,02	0,78	0,03	0,78	0,03
Yeast1vs7	0,73	0,15	0,77	0,09	0,56	0,28	0,68	0,09	0,61	0,16	0,72	0,05	0,81	0,06	0,84	0,08
Yeast1289vs7	0,73	0,19	0,77	0,15	0,36	0,08	0,68	0,09	0,50	0,03	0,72	0,07	0,64	0,13	0,79	0,08
Yeast1458vs7	0,60	0,19	0,77	0,22	0,40	0,05	0,58	0,07	0,48	0,08	0,66	0,08	0,59	0,18	0,67	0,09
Yeast2vs4	0,74	0,42	0,92	0,13	0,67	0,20	0,73	0,41	0,59	0,34	0,72	0,41	0,94	0,04	0,98	
Yeast2vs8	0,80	0,11	0,60	0,14	0,48	0,24	0,87	0,09	0,60	0,14	0,72	0,10	0,75	0,12	0,81	0,08
Yeast3	0,92	0,04	0,92	0,03	0,75	0,13	0,91	0,02	0,83	0,06	0,91	0,03	0,96	0,01	0,97	0,01
Yeast4	0,86	0,05	0,86	0,09	0,67	0,16	0,71	0,21	0,76	0,08	0,77	0,10	0,86	0,08	0,90	0,03
Yeast5	1,00	0,00	1,00	0,00	0,64	0,05	0,94	0,03	0,80	0,03	0,97	0,01	0,99	0,01	0,99	0,00
Yeast6	0,91	0,13	0,89	0,16	0,59	0,08	0,91	0,03	0,73	0,07	0,89	0,07	0,93	0,08	0,93	0,09
Large scale datasets																
Cancer	0,83	0,04	0,87	0,04	0,78	0,01	0,81	0,01	0,79	0,03	0,84	0,02	0,91	0,02	0,93	0,01
Protein	0,93	0,02	0,97	0,02	0,90	0,01	0,93	0,01	0,90	0,01	0,95	0,01	0,98	0,01	0,99	0,00
Avg.	0,89	0,11	0,87	0,11	0,74	0,17	0,84	0,15	0,80	0,14	0,84	0,12	0,90	0,10	0,92	0,09

small λ values, the classification performance varies, but after $\lambda = 30$, the AUC value stabilizes. For an in-depth illustration of the λ impact, we performed an additional experiment over the “two moons” synthetic dataset. As we can see in Fig. 4, the larger the lambda value, the more scattered the points are located. In short, as the value of λ is increased, more emphasis is given to the similarity between X_+ and X_- . Accordingly, from the results in Fig. 3, 4, for further experiments we fix $\lambda = 100$.

In turn, Table 2 exhibits the classification performance of RIUS and CRIUS approaches for the real-world datasets concerning sensitivity, specificity, G-mean, and AUC. Besides, Table 3 shows the undersampling time in seconds and the average iterations per fold. The best results are highlighted in bold. Overall, we notice that CRIUS achieves the highest classification performance in most of the evaluation metrics. However, RIUS gets a remarkable average sensitivity performance. Although RIUS exhibits an outstanding

sensitivity in small scale datasets, CRIUS exceeds RIUS in all measurements in large scale databases. Additionally, CRIUS requires less computational time and a lower number of iterations for every single dataset. As seen, for the large scale datasets, the undersampling process decreases from hours to minutes. The latter can be explained due to CRIUS is a highly parallelizable method; therefore, it can be performed over each cluster independently, saving time and computational resources.

Table 3 also shows how the Mparameter of CRIUS affects the undersampling time and the number of iterations until the method's convergence. Three different values of M were tested, namely $M = \{3, 4, 5\}$. The elapsed time by CRIUS with the three values of M are similar. Of note, M determines the number of classifiers to the ensemble and the imbalanced ratio ($M : 1$). We decided to fix $M = 3$ as a trade-off between classification performance and undersampling time for further experiments.

Table 3

Computational time results (RIUS vs. CRIUS). The mean $\pm$ standard deviation are depicted concerning the undersampling time in seconds for each fold. The average iterations per fold are also reported in brackets. Bold: Best results per database.

Dataset	RIUS		CRIUS					
			M = 3		M = 4		M = 5	
Small scale datasets								
Abalone9-18	1,85	0,15 (73)	0,74	0,04 (39)	0,74	0,14 (37)	0,52	0,07 (26)
Abalone19	3,96	0,47 (120)	1,62	0,14 (82)	1,48	0,12 (75)	0,97	0,16 (45)
Ecoli-0_vs_1	6,75	1,53 (304)	0,7	0,09 (34)	0,68	0,11 (34)	0,49	0,1 (24)
Ecoli-0-1-3-7_vs_2-6	9,57	0,99 (673)	0,58	0,11 (28)	0,57	0,09 (29)	0,57	0,11 (30)
Ecoli1	10,09	1,17 (436)	0,74	0,07 (37)	0,72	0,09 (36)	0,66	0,05 (33)
Ecoli2	9,59	0,83 (489)	0,74	0,08 (31)	0,77	0,04 (34)	0,72	0,06 (31)
Ecoli3	9,08	2,37 (527)	0,4	0,06 (27)	0,43	0,04 (30)	0,42	0,04 (30)
Ecoli4	10,92	1,95 (663)	0,4	0,04 (29)	0,36	0,04 (26)	0,31	0,04 (23)
Glass0	8,07	0,74 (373)	0,72	0,04 (51)	0,71	0,04 (51)	0,63	0,04 (44)
Glass0123vs456	7,56	1,21 (409)	0,39	0,04 (25)	0,63	0,05 (43)	0,55	0,03 (37)
Glass016vs2	9,1	1,12 (626)	0,51	0,08 (35)	0,58	0,09 (40)	0,42	0,15 (29)
Glass016vs5	10,12	1,9 (694)	0,47	0,14 (31)	0,61	0,08 (41)	0,44	0,13 (28)
Glass1	8,02	1,1 (377)	0,94	0,19 (50)	0,97	0,15 (51)	0,9	0,16 (44)
Glass2	10,58	0,69 (659)	0,58	0,05 (38)	0,58	0,12 (40)	0,46	0,05 (32)
Glass4	9,13	0,7 (651)	0,5	0,09 (33)	0,59	0,06 (41)	0,51	0,09 (35)
Glass5	13,23	5,07 (881)	0,49	0,1 (30)	0,67	0,12 (43)	0,51	0,17 (34)
Glass6	8,73	0,61 (523)	0,47	0,06 (32)	0,57	0,04 (41)	0,44	0,06 (31)
Haberman	47,89	17,19 (1184)	0,61	0,1 (37)	0,41	0,12 (22)	0,31	0,07 (17)
Iris0	43,36	12,81 (1342)	0,38	0,08 (21)	0,41	0,08 (23)	0,32	0,06 (18)
New-thyroid1	25,33	8,67 (783)	0,37	0,07 (23)	0,28	0,05 (18)	0,28	0,05 (18)
New-thyroid2	15,08	1,89 (767)	0,36	0,01 (21)	0,32	0,03 (18)	0,31	0,06 (18)
Page-blocks0	706,9	71,33 (794)	1,3	0,06 (57)	1,27	0,04 (53)	1,13	0,06 (49)
Page-blocks13vs4	34,48	5,79 (1501)	1,15	0,44 (60)	0,98	0,3 (55)	1,04	0,47 (49)
Pima	4,52	0,79 (89)	1,03	0,26 (44)	0,78	0,33 (38)	0,75	0,22 (38)
Segment0	56,2	2,72 (43)	1,64	0,22 (55)	2,37	1,69 (56)	1,66	0,39 (55)
Shuttle0vs4	52,5	13,11 (991)	1,76	0,34 (70)	1,69	0,66 (65)	1,51	0,47 (60)
Shuttle2vs4	8,14	2,19 (411)	1,8	0,35 (49)	2,06	0,39 (57)	1,78	0,34 (50)
Vehicle0	7,55	2,02 (102)	1,55	0,23 (56)	1,9	0,08 (63)	1,86	0,09 (62)
Vehicle1	5,78	0,95 (103)	1,83	0,15 (56)	1,98	0,05 (60)	2	0,15 (60)
Vehicle2	7,03	2,09 (75)	1,41	0,35 (58)	1,54	0,38 (63)	1,69	0,38 (62)
Vehicle3	5,11	0,24 (103)	1,55	0,1 (56)	1,74	0,02 (61)	1,71	0,04 (60)
Vowel0	7,37	1,11 (266)	0,96	0,19 (38)	1,06	0,12 (39)	1,06	0,09 (38)
Wisconsin	13,79	1,65 (359)	0,09	0 (2)	0,87	0,09 (28)	0,9	0,06 (28)
Yeast05679vs4	2,97	0,32 (165)	0,73	0,12 (23)	0,77	0,12 (25)	0,68	0,13 (21)
Yeast1	19,85	1,93 (198)	1,15	0,1 (38)	1,03	0,13 (37)	0,94	0,15 (35)
Yeast1vs7	3,36	0,28 (173)	0,44	0,08 (22)	0,44	0,13 (22)	0,52	0,08 (25)
Yeast1289vs7	4,58	0,39 (260)	0,46	0,08 (25)	0,41	0,03 (24)	0,44	0,09 (25)
Yeast1458vs7	4,3	0,74 (221)	0,43	0,09 (23)	0,39	0,05 (21)	0,46	0,07 (23)
Yeast2vs4	4,93	1,43 (189)	0,73	0,17 (24)	0,82	0,2 (28)	0,78	0,18 (27)
Yeast2vs8	5,39	1,14 (207)	0,77	0,22 (25)	0,64	0,08 (22)	0,76	0,14 (25)
Yeast3	13,68	0,79 (229)	0,68	0,1 (24)	0,72	0,1 (25)	0,72	0,07 (24)
Yeast4	6,5	0,18 (238)	0,69	0,16 (23)	0,65	0,21 (20)	0,72	0,2 (23)
Yeast5	6,41	0,31 (234)	0,83	0,16 (24)	0,75	0,25 (21)	0,75	0,26 (22)
Yeast6	6,33	0,42 (243)	0,98	0,13 (25)	0,8	0,05 (22)	0,84	0,16 (22)
Large scale datasets								
Cancer	27880,6	788,2 (24870)	72,15	0,93 (201)	72,45	1,03 (208)	72,63	0,99 (213)
Protein	46680,8	1352,1 (38655)	112,03	0,88 (180)	112,13	0,97 (184)	112,77	0,78 (200)

5.2. Method comparison results

Table 4 shows the method comparison results over the real-world datasets. The AUC is reported to make the results comparable according to the state-of-the-art. As shown, the proposed CRIUS and RIUS demonstrated the highest and second-highest classification performance. For small scale datasets, compared with the four state-of-the-art approaches, our RIUS method wins in the 56.8% of the cases. Besides, RIUS combined with the clustering procedure and the bagging ensemble (CRIUS), wins 70.45% of the evidence. It is essential to highlight that the RIUS approach is just a subsampling method, and its classification results were achieved by training only one classifier. However, RIUS was superior to RUS1, UB4, SBAG4, and CUS-AB, which are ensemble methods that require a higher number of classifiers.

Table 4 also outlines the achieved performances for the large-scale datasets. As seen, for both datasets, the CRIUS approach exhibits the highest AUC. The RIUS approach reaches similar performances in comparison with the CUS-Adaboost methodology. The

latter can be explained because RIUS places more points on regions where the PDF estimated from the majority class has large values. This property is highly problem dependent, but the RIUS approach located more points on the densest regions for the large-scale datasets. In these scenarios, the CRIUS and CUS-AB achieve the highest performances since clustering the dataset into $K = N_{-}$ groups allows them to place points in all the feature space regions. Nevertheless, CUS-AB selects each cluster centroid, which is problematic because the majority class's shape and outliers affect the mean value, rendering the group center entirely meaningless. Then, the point selected by CUS-AB does not represent the main structure of the group properly. In contrast, since RIUS and CRIUS employ a Renyi's quadratic entropy-based functional, such a higher-order statistics expressed by ITL allows extracting the information from each cluster to suit the relevant structural properties within each class.

Finally, we quantify the statistical differences between our proposals and the state-of-the-art methods concerning the AUC performance through the paired Wilcoxon signed-rank test

Table 4

Method comparison results. KEEL (small scale datasets) and KDDMC (large scale dataset) repositories are studied. The AUC of the testing set is presented. NP: Not presented.

Dataset	IR	State of the art approaches				Our approaches	
		RUS1[28]	UB4[29]	SBAG4[30]	CUS-AB[10]	RIUS	CRIUS
<i>Small scale datasets</i>							
Abalone9-18	16,68	0,693	0,719	0,745	0,831	0,893	0,955
Abalone19	128,87	0,631	0,721	0,572	0,728	0,812	0,824
Ecoli-0_vs_1	1,86	0,969	0,98	0,983	0,982	0,984	1,000
Ecoli-0-1-3-7_vs_2-6	39,15	0,794	0,745	0,828	0,804	0,836	0,946
Ecoli1	3,36	0,883	0,9	0,9	0,927	0,932	0,951
Ecoli2	5,46	0,899	0,884	0,888	0,947	0,956	0,944
Ecoli3	8,19	0,856	0,908	0,885	0,926	0,939	0,942
Ecoli4	13,84	0,942	0,888	0,933	0,95	0,984	0,985
Glass0	3,19	0,813	0,814	0,839	0,873	0,879	0,867
Glass0123vs456	10,29	0,93	0,904	0,946	0,97	0,965	0,979
Glass016vs2	19,44	0,617	0,754	0,559	0,79	0,795	0,838
Glass016vs5	1,82	0,989	0,943	0,866	0,964	0,906	0,963
Glass1	10,39	0,763	0,737	0,728	0,824	0,712	0,812
Glass2	15,47	0,78	0,769	0,779	0,76	0,859	0,815
Glass4	22,81	0,915	0,846	0,874	0,853	0,937	0,967
Glass5	22,81	0,943	0,949	0,878	0,949	0,909	0,978
Glass6	6,38	0,918	0,904	0,931	0,905	0,974	0,978
Haberman	2,68	0,655	0,664	0,656	0,603	0,693	0,603
Iris0	2	0,99	0,99	0,98	0,99	1,000	1,000
New-thyroid1	5,14	0,958	0,964	0,975	0,973	0,987	0,999
New-thyroid2	4,92	0,938	0,958	0,961	0,924	0,983	0,997
Page-blocks0	8,77	0,948	0,958	0,953	0,986	0,974	0,984
Page-blocks13vs4	15,85	0,987	0,978	0,988	0,992	0,991	0,995
Pima	1,9	0,726	0,76	0,751	0,758	0,815	0,805
Segment0	6,01	0,993	0,988	0,994	0,996	0,999	1,000
Shuttle0vs4	13,87	1	1	1	1	1,000	1,000
Shuttle2vs4	20,5	1	1	1	0,988	1,000	1,000
Vehicle0	3,23	0,958	0,952	0,965	0,99	0,990	0,998
Vehicle1	2,52	0,747	0,787	0,769	0,832	0,907	0,917
Vehicle2	2,52	0,97	0,964	0,966	0,995	0,993	0,997
Vehicle3	2,52	0,765	0,802	0,763	0,827	0,874	0,912
Vowel0	10,1	0,943	0,947	0,988	0,987	0,999	1,000
Wisconsin	1,86	0,964	0,96	0,96	0,99	0,980	0,991
Yeast05679vs4	9,35	0,803	0,794	0,818	0,869	0,833	0,870
Yeast1	2,46	0,719	0,722	0,734	0,747	0,777	0,783
Yeast1vs7	13,87	0,715	0,786	0,697	0,768	0,808	0,836
Yeast1289vs7	30,56	0,721	0,734	0,658	0,692	0,638	0,793
Yeast1458vs7	22,1	0,567	0,606	0,623	0,627	0,592	0,671
Yeast2vs4	9,08	0,933	0,936	0,897	0,977	0,938	0,979
Yeast2vs8	23,1	0,789	0,783	0,784	0,868	0,753	0,809
Yeast3	8,11	0,925	0,934	0,944	0,967	0,961	0,971
Yeast4	28,41	0,812	0,855	0,773	0,874	0,859	0,897
Yeast5	32,78	0,959	0,952	0,962	0,987	0,987	0,986
Yeast6	39,15	0,823	0,869	0,836	0,909	0,928	0,930
<i>Large scale datasets</i>							
Cancer	163	NP	0,871	NP	0,921	0,910	0,932
Protein	111	NP	0,953	NP	0,980	0,979	0,992
Avg.		0,856	0,866	0,853	0,891	0,900	0,922
wins		1/46	0/46	0/46	4/46	7/46	34/46

Table 5

Wilcoxon signed-rank test results. The p-value is shown to compare the AUC values presented in Table 2. Bold: p-value < 0.05 (the AUC along the studied databases differs significantly).

	RUS1[28]	UB4[29]	SBAG4[30]	CUS-AB[10]	RIUS (ours)	CRIUS (ours)
RUS1[28]	–	0.218	0.731	0.000	0.000	0.000
UB4[29]	0.218	–	0.824	0.000	0.000	0.000
SBAG4[30]	0.731	0.824	–	0.000	0.000	0.000
CUS-AB[10]	0.000	0.000	0.000	–	0.110	0.000
RIUS (ours)	0.000	0.000	0.000	0.110	–	0.000
CRIUS (ours)	0.000	0.000	0.000	0.000	0.000	–

[31]. All significance levels are measured at 5%. Therefore, if we observe a p-value > 0.05, then the null hypothesis that there is no high difference between AUC values can not be rejected; otherwise, if p-value ≤ 0.05, then the null hypothesis is rejected. As seen in.

Table 5, RIUS is statistically superior to RUS1, UB4, and SBAG4. Besides, CRIUS is statistically superior to RUS1, UB4, SBAG4, RIUS, and CUS-AB. Overall, CRIUS is the best trade-off between classification performance and the required computational time for small and large scale datasets.

6. Conclusions

This paper presents a new undersampling technique called Relevant Information UnderSampling-(RIUS), to support imbalanced data binary classification tasks. To this end, RIUS minimizes the entropy of M points properly initialized while preserving different levels of detail about the majority class through a Cauchy Schwarz-based divergence loss. Moreover, our RIUS approach quantifies the majority class's relevant structure beyond second-order statistics through a Renyi's quadratic-based entropy, making the method robust to complex data distributions. We also propose a hybrid approach that combines a clustering-based undersampling and a bagging ensemble method. The new variation, named CRIUS, produces K clusters over the majority class by the K-means algorithm. Here, the number of clusters K is set to be equal to the number of data samples in the minority class ($K = N_-$). We then run the RIUS algorithm but select as many samples in each cluster as classifiers want to combine. Finally, we build balanced training sets to train SVM-based discriminants. We tested our approaches on synthetic datasets for illustrative purposes and free parameter analysis. Also, 44 small-scale binary datasets from the KEEL dataset repository and two large-scale databases from the KDDMC are used for real-world data testing. The results show that the proposed methods are competitive in terms of classification performance and computational time. In fact, in most cases, our approaches achieve better results, when compared to different state-of-the-art approaches [28–30,10]. The above is explained because RIUS and CRIUS subsample the majority class without losing the underlying structure.

As future work, we want to formulate a subsampling approach that considers the minority class to alleviate the overlapping between clusters. Besides, the authors plan to extend the RIUS and CRIUS methods for multi-class tasks. We want to include a redundancy criterion to weigh the subsampled points concerning the dataset's local densities; that is, to obtain fewer samples around high-density neighborhoods, especially for significant scale problems. Finally, we plan to improve the scalability of the methods using stochastic gradient descent and mini-batch techniques.

CRediT authorship contribution statement

J. Hoyos-Osorio: Methodology, Investigation, Software, Writing - original draft. **A. Alvarez-Meza:** Conceptualization, Formal analysis, Writing - original draft, Writing - review & editing. **G. Daza-Santacoloma:** Conceptualization, Data curation, Resources. **A. Orozco-Gutierrez:** Supervision, Project administration, Funding acquisition. **G. Castellanos-Domínguez:** Writing - review & editing, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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